# Ejercicio 4 (8): Gestión de impresoras e impresión - Gestión de la hora del sistema

**NOTA:** Este ejercicio se puede completar usando cualquier máquina virtual

Adjuntar capturas de pantalla para ver los resultados

## PARTE I - Impresión

1. Instale el servicio de impresión CUPS y LPD legacy.

apt-get install cups lpr

```
vagrant@debian:~$ sudo apt-get install cups lpr
sudo: unable to resolve host debian: Name or service not known
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme at-spi2-core fontconfig giblib1 glib-networking glib-networking-common glib-networking-services gnome-icon-theme gsettings-desktop-schemas
  gtk-update-icon-cache hicolor-icon-theme libatk-bridge2.0-0 libatk1.0-0 libatk1.0-data libatspi2.0-0 libcairo-gobject2 libcroco3 libdatrie1 libgail-common libgail18
  libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgraphite2-3 libgtk-3-0 libgtk-3-bin libgtk-3-common libgtk2.0-0 libgtk2.0-bin libgtk2.0-common
  libharfbuzz0b libid3tag0 libimlib2 libjson-glib-1.0-0 libjson-glib-1.0-common libobrender32v5 libobt2v5 libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpangoxft-1.0-0
  libproxy1v5 librest-0.7-0 librsvg2-2 librsvg2-common libsoup-gnome2.4-1 libsoup2.4-1 libstartup-notification0 libthai-data libthai0 libwayland-cursor0 libwayland-egl1
  libxkbcommon0 obconf obsession scrot
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  acl avahi-daemon bc colord colord-data cups-browsed cups-client cups-common cups-core-drivers cups-daemon cups-filters cups-filters-core-drivers cups-ipp-utils cups-ppdc
  cups-server-common fonts-droid-fallback fonts-noto-mono ghostscript gsfonts libavahi-core7 libavahi-glib1 libcolorhug2 libcupsfilters1 libcupsimage2 libexif12 libfontembed1
  libgd3 libgphoto2-6 libgphoto2-l10n libgphoto2-port12 libgs9 libgs9-common libgusb2 libgutenprint-common libgutenprint9 libieee1284-3 libijs-0.35 libjbig2dec0 liblouis-data
  liblouis17 liblouisutdml-bin liblouisutdml-data liblouisutdml8 libnspr4 libnss-mdns libnss3 libopenjp2-7 libpaper-utils libpaper1 libpolkit-agent-1-0 libpolkit-backend-1-0
  libpolkit-gobject-1-0 libpoppler82 libqpdf21 libsane libsane-common libsnmp-base libsnmp30 libusb-1.0-0 libyaml-0-2 policykit-1 poppler-data poppler-utils
  printer-driver-gutenprint sane-utils ssl-cert update-inetd
Suggested packages:
  colord-sensor-argyll cups-bsd foomatic-db-compressed-ppds | foomatic-db printer-driver-hpcups hplip cups-pdf smbclient antiword docx2txt imagemagick fonts-noto ghostscript-x
  libgd-tools gphoto2 gutenprint-locales ooo2dbk rtf2xml snmp-mibs-downloader magicfilter | apsfilter fonts-japanese-mincho | fonts-ipafont-mincho fonts-japanese-gothic
  | fonts-ipafont-gothic fonts-arphic-ukai fonts-arphic-uming fonts-nanum gutenprint-doc unpaper openssl-blacklist
```

captuara inicial

Modificamos el fichero de configuración “nano /etc/cups/cupsd.conf” y modificamos los siguientes parámetos; modificamos lo siguiente (en rojo las modificaciones), permitiendo las conexiones remotas y el acceso para todo el munod tanto al servidor como a las páginas de administración.

```
# Only listen for connections from the local machine.
#Listen localhost:631
Listen 631

# Restrict access to the server...
<Location />
  Order allow,deny
  Allow from all
</Location>

# Restrict access to the admin pages...
<Location /admin>
  Order allow,deny
  Allow from all
</Location>
```

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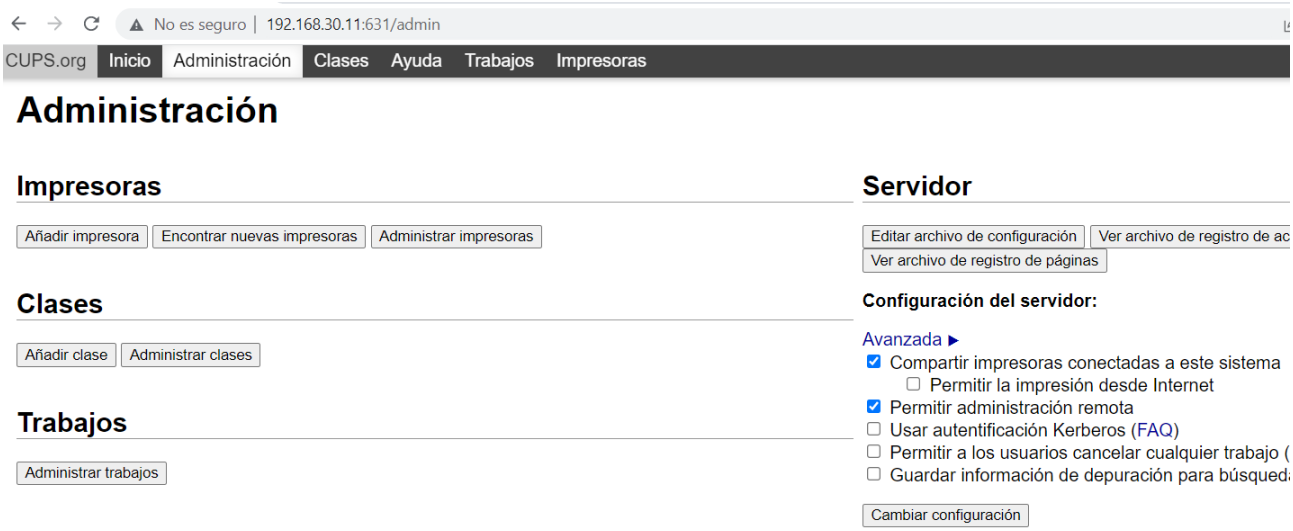
```
  GNU nano 3.2                                                    /etc/

#
# Configuration file for the CUPS scheduler.  See "man cupsd.conf" for a
# complete description of this file.
#

# Log general information in error_log - change "warn" to "debug"
# for troubleshooting...
LogLevel warn
PageLogFormat

# Deactivate CUPS' internal logrotating, as we provide a better one, especially
# LogLevel debug2 gets usable now
MaxLogSize 0

# Only listen for connections from the local machine.
#Listen localhost:631
Listen 631
Listen /run/cups/cups.sock

# Show shared printers on the local network.
Browsing On
BrowseLocalProtocols dnssd

# Default authentication type, when authentication is required...
DefaultAuthType Basic

# Web interface setting...
WebInterface Yes

# Restrict access to the server...
<Location />
  Order allow,deny
  Allow from all
</Location>

# Restrict access to the admin pages...
<Location /admin>
  Order allow,deny
  Allow from all
</Location>
```

“Systemctl restart cups.service” y comprobamos que accedemos a la impresora por web ip del equipo y puerto 631 (http://192.168.30.11:631/) y accedemos a la página de configuración



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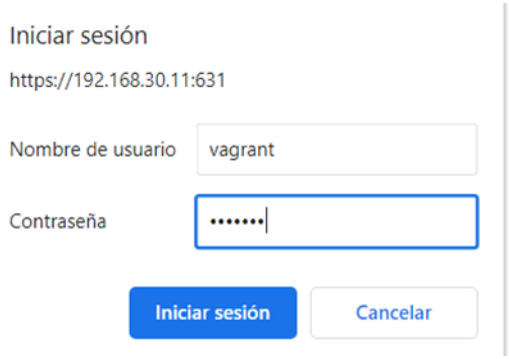
Añadimos el usuario vagrant a lpadmin “gpasswd -a vagrant lpadmin” (para poder administrar cups con el usuario vagrant

```
root@debian:/home/vagrant# gpasswd -a vagrant lpadmin
Adding user vagrant to group lpadmin
root@debian:/home/vagrant#
```



Iniciamos sesión con el usuario
Y ya podemos agregar una impresora

2. Instale la impresora CUPS-PDF.

sudo apt-get isntall cups-pdf

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```
root@debian:/home/vagrant# sudo apt-get install cups-pdf
sudo: unable to resolve host debian: Name or service not known
Reading package lists... Done
Building dependency tree
Reading state information... Done
Note, selecting 'printer-driver-cups-pdf' instead of 'cups-pdf'
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme at-spi2-core fontconfig giblib1 glib-networking glib-networking-common
  glib-networking-services gnome-icon-theme gsettings-desktop-schemas gtk-update-icon-cache
  hicolor-icon-theme libatk-bridge2.0-0 libatk1.0-0 libatk1.0-data libatspi2.0-0 libcairo-gobject2 libcroco3
  libdatrie1 libgail-common libgail18 libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7
  libgraphite2-3 libgtk-3-0 libgtk-3-bin libgtk-3-common libgtk2.0-0 libgtk2.0-bin libgtk2.0-common
  libharfbuzz0b libid3tag0 libimlib2 libjson-glib-1.0-0 libjson-glib-1.0-common libobrender32v5 libobt2v5
  libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpangoxft-1.0-0 libproxy1v5 librest-0.7-0 librsvg2-2
  librsvg2-common libsoup-gnome2.4-1 libsoup2.4-1 libstartup-notification0 libthai-data libthai0
  libwayland-cursor0 libwayland-egl1 libxkbcommon0 obconf obsession scrot
Use 'sudo apt autoremove' to remove them.
Suggested packages:
  system-config-printer
The following NEW packages will be installed:
  printer-driver-cups-pdf
0 upgraded, 1 newly installed, 0 to remove and 19 not upgraded.
Need to get 36.9 kB of archives.
```

Vamos a la web de cups, en la parte  de administración, vamos a añadir una impreosra y nos aparece el menú, en elque figura la impresora seleccionamos y pulsamos en siguiente:

pulsamos en siguiente:

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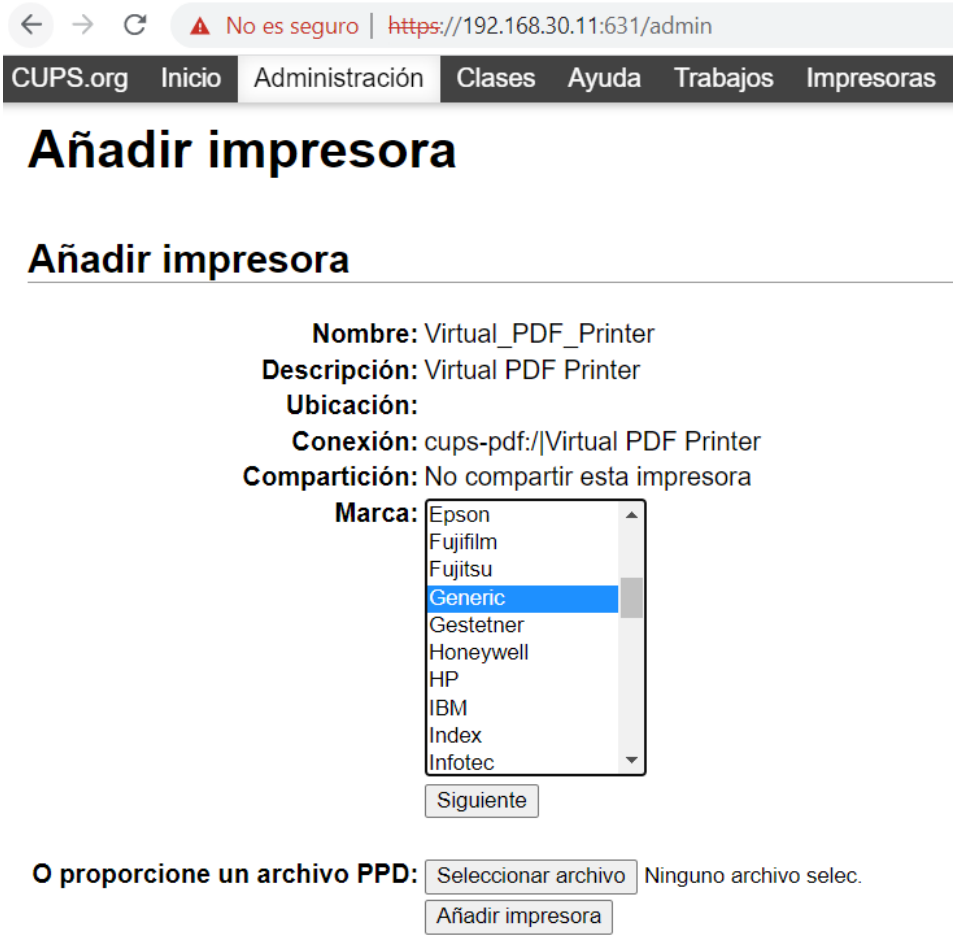
seleccionamos generic



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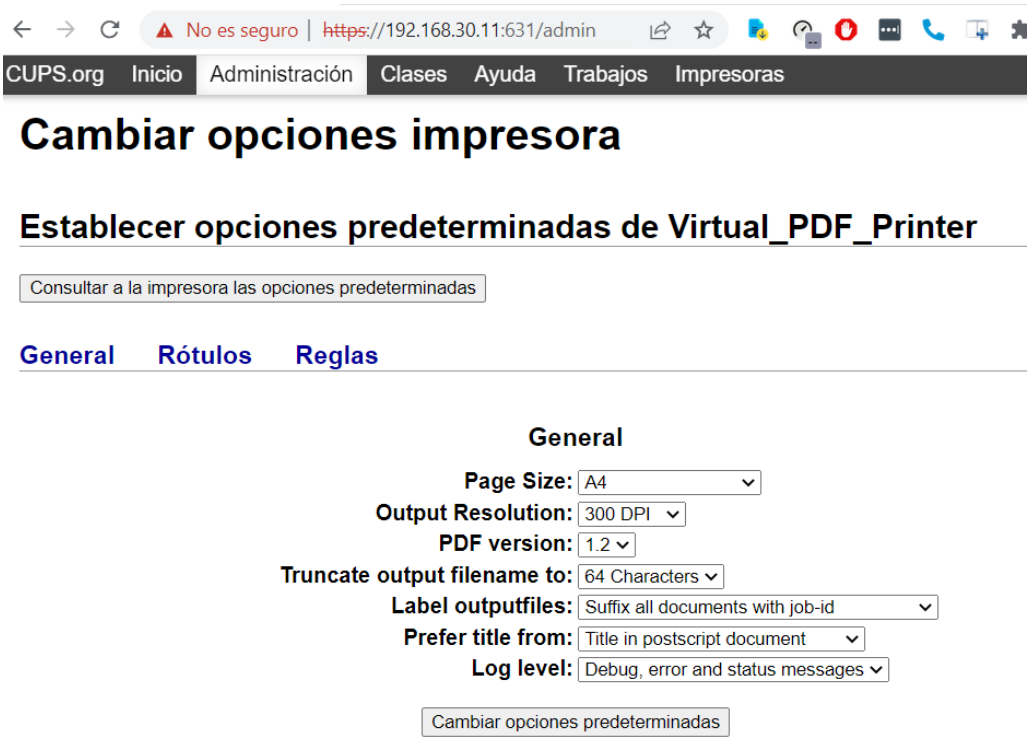
Pulsamos en añadir impresora y ya tenemos la impresora instalada.



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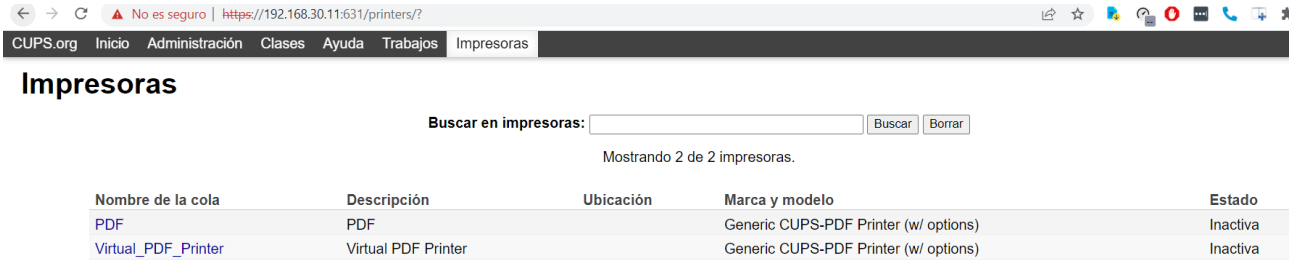
Si vamos a impresoras vemos que hemos añadido una nueva impresora pdf




3. Inicie el servicio CUPS, verifique que se esté ejecutando.

systemctl start cups.service
systemctl status cups.service

```
root@debian:/home/vagrant# systemctl start cups.service
root@debian:/home/vagrant# systemctl status cups.service
● cups.service - CUPS Scheduler
   Loaded: loaded (/lib/systemd/system/cups.service; enabled; vendor preset: enabled)
   Active: active (running) since Fri 2023-04-21 03:29:06 UTC; 27min ago
     Docs: man:cupsd(8)
 Main PID: 3980 (cupsd)
    Tasks: 3 (limit: 1146)
   Memory: 6.1M
   CGroup: /system.slice/cups.service
           ├─3980 /usr/sbin/cupsd -l
           └─4013 /usr/lib/cups/notifier/dbus dbus://

Apr 21 03:29:06 debian systemd[1]: Started CUPS Scheduler.
root@debian:/home/vagrant#
```

4. Enumere todas las impresoras junto con el conjunto de impresoras predeterminado.

lpstat -s

```
root@debian:/home/vagrant# lpstat -s
destino predeterminado del sistema: PDF
dispositivo para PDF: cups-pdf:/
dispositivo para Virtual_PDF_Printer: cups-pdf:/
root@debian:/home/vagrant#
```

5. Configure la impresora CUPS-PDF como predeterminada.

lpoptions -d Virtual_PDF_Printer

Lpstat -s (para comprobar)

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```
root@debian:/home/vagrant# lpoptions -d Virtual_PDF_Printer
copies=1 device-uri=cups-pdf:/ finishings=3 job-cancel-after=10800 job-hold-until=no-hold job-priority=50 job-s
heets=none,none marker-change-time=0 number-up=1 printer-commands=AutoConfigure,Clean,PrintSelfTestPage printer
-info='Virtual PDF Printer' printer-is-accepting-jobs=true printer-is-shared=false printer-is-temporary=false p
rinter-location printer-make-and-model='Generic CUPS-PDF Printer (w/ options)' printer-state=3 printer-state-ch
ange-time=1682048960 printer-state-reasons=none printer-type=10547276 printer-uri-supported=ipp://localhost/pri
nters/Virtual_PDF_Printer
root@debian:/home/vagrant# lpstat -s
destino predeterminado del sistema: Virtual_PDF_Printer
dispositivo para PDF: cups-pdf:/
dispositivo para Virtual_PDF_Printer: cups-pdf:/
root@debian:/home/vagrant#
```

6. Verifique que se muestre como predeterminado y muestre las opciones de la impresora para esa impresora

lpoptions
lpoptions -l

```
root@debian:/home/vagrant# lpoptions
copies=1 device-uri=cups-pdf:/ finishings=3 job-cancel-after=10800 job-hold-until=no-hold job-priority=50 job-s
heets=none,none marker-change-time=0 number-up=1 printer-commands=AutoConfigure,Clean,PrintSelfTestPage printer
-info='Virtual PDF Printer' printer-is-accepting-jobs=true printer-is-shared=false printer-is-temporary=false p
rinter-location printer-make-and-model='Generic CUPS-PDF Printer (w/ options)' printer-state=3 printer-state-ch
ange-time=1682048960 printer-state-reasons=none printer-type=10547276 printer-uri-supported=ipp://localhost/pri
nters/Virtual_PDF_Printer
root@debian:/home/vagrant# lpoptions -l
PageSize/Page Size: Custom.WIDTHxHEIGHT 11x14 11x17 13x19 16x20 16x24 2A 4A 8x10 8x12 A0 A1 A2 A3 *A4 A5 AnsiA
AnsiB AnsiC AnsiD AnsiE ArchA ArchB ArchC ArchD ArchE C0 C1 C2 C3 C4 C5 Env10 EnvC5 EnvDL EnvMonarch Executive
ISOB0 ISOB1 ISOB2 ISOB3 ISOB4 ISOB5 JISB0 JISB1 JISB2 JISB3 JISB4 JISB5 Ledger Legal Letter RA0 RA1 RA2 RA3 RA4
 SRA0 SRA1 SRA2 SRA3 SRA4 SuperA SuperB TabloidExtra Tabloid
Resolution/Output Resolution: 150dpi *300dpi 600dpi 1200dpi 2400dpi
PDFVer/PDF version: 1.1 *1.2 1.3 1.4 1.5
Truncate/Truncate output filename to: 8 16 32 *64
Label/Label outputfiles: 0 1 *2
TitlePref/Prefer title from: *0 1
LogType/Log level: 1 3 *7
root@debian:/home/vagrant#
```

En la web también se puede ver las opciones predeterminadas

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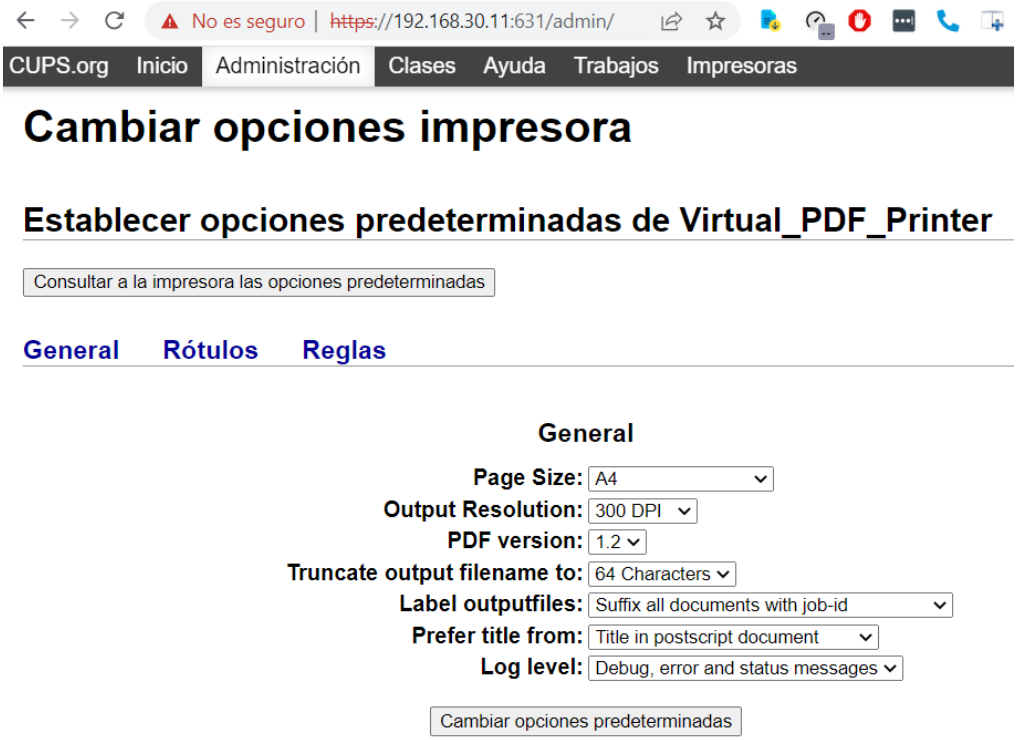


### PARTE II - hora del sistema

1. Muestre la fecha y hora actuales en el sistema. Muestra la hora equivalente a UTC. Adelante en una hora la hora del sistema.

   Con “date” comprobamos la hora y con date –set “2023-04-21 05:16”

```
root@debian:/home/vagrant# date
Fri Apr 21 04:16:37 UTC 2023
root@debian:/home/vagrant# date --set "2023-04-21 05:16"
Fri Apr 21 05:16:00 UTC 2023
root@debian:/home/vagrant# date
Fri Apr 21 05:16:01 UTC 2023
root@debian:/home/vagrant#
```

2. Muestre la hora del reloj de hardware del sistema. Restablece la hora del sistema en función de a hora del reloj de hardware.

   hwclock
   hwclock --systoch

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```
root@debian:/home/vagrant# hwclock
2023-04-21 04:08:43.429563+00:00
root@debian:/home/vagrant# sudo hwclock --systohc
sudo: unable to resolve host debian: Name or service not known
root@debian:/home/vagrant# date
Fri Apr 21 05:19:02 UTC 2023
root@debian:/home/vagrant# hwclock
2023-04-21 05:19:26.136283+00:00
root@debian:/home/vagrant#
```

3. Enumera el directorio que es responsable de toda la información de las zonas horarias.

ls /usr/share/zoneinfo

```
root@debian:/usr/share/zoneinfo# ls /usr/share/zoneinfo/
Africa      CET      Egypt    GMT+0      Iran       MST7MDT  Poland     US                localtime
America     CST6CDT  Eire     GMT-0      Israel     Mexico   Portugal   UTC               posix
Antarctica  Canada   Etc      GMT0       Jamaica    NZ       ROC        Universal         posixrules
Arctic      Chile    Europe   Greenwich  Japan      NZ-CHAT  ROK        W-SU              right
Asia        Cuba     Factory  HST        Kwajalein  Navajo   Singapore  WET               zone.tab
Atlantic    EET      GB       Hongkong   Libya      PRC      SystemV    Zulu              zone1970.tab
Australia   EST      GB-Eire  Iceland    MET        PST8PDT  Turkey     iso3166.tab
Brazil      EST5EDT  GMT      Indian     MST        Pacific  UCT        leap-seconds.list
root@debian:/usr/share/zoneinfo#
```

4. **Instale el servicio ntp.** Busque servidores públicos **NTP** para su región del mundo / país y agré-guelos a su configuración. Reinicie el servicio NTP.

apt install ntp

```
root@debian:/usr/share/zoneinfo# apt install ntp
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following packages were automatically installed and are no longer required:
  adwaita-icon-theme at-spi2-core fontconfig giblib1 glib-networking glib-networking-common
  glib-networking-services gnome-icon-theme gsettings-desktop-schemas gtk-update-icon-cache
  hicolor-icon-theme libatk-bridge2.0-0 libatk1.0-0 libatk1.0-data libatspi2.0-0 libcairo-gobject2 libcroco3
  libdatrie1 libgail-common libgail18 libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7
  libgraphite2-3 libgtk-3-0 libgtk-3-bin libgtk-3-common libgtk2.0-0 libgtk2.0-bin libgtk2.0-common
  libharfbuzz0b libid3tag0 libimlib2 libjson-glib-1.0-0 libjson-glib-1.0-common libobrender32v5 libobt2v5
  libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpangoxft-1.0-0 libproxy1v5 librest-0.7-0 librsvg2-2
  librsvg2-common libsoup-gnome2.4-1 libsoup2.4-1 libstartup-notification0 libthai-data libthai0
  libwayland-cursor0 libwayland-egl1 libxkbcommon0 obconf obsession scrot
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libevent-core-2.1-6 libevent-pthreads-2.1-6 libopts25 sntp
Suggested packages:
```

Editamos el fichero “nano /etc/ntp.conf”, comentamos los servidores y añadimos los nuevos.

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```
  GNU nano 3.2                                        /etc/ntp.conf

filegen clockstats file clockstats type day enable


# You do need to talk to an NTP server or two (or three).
#server ntp.your-provider.example

# pool.ntp.org maps to about 1000 low-stratum NTP servers.  Your server will
# pick a different set every time it starts up.  Please consider joining the
# pool: <http://www.pool.ntp.org/join.html>
#pool 0.debian.pool.ntp.org iburst
#pool 1.debian.pool.ntp.org iburst
#pool 2.debian.pool.ntp.org iburst
#pool 3.debian.pool.ntp.org iburst
server 0.es.pool.ntp.org
server 1.es.pool.ntp.org
server 0.europe.pool.ntp.org
server 1.europe.pool.ntp.org


# Access control configuration; see /usr/share/doc/ntp-doc/html/accopt.html for
# details.  The web page <http://support.ntp.org/bin/view/Support/AccessRestrictions>
# might also be helpful.
```

Systemctl restart ntp

```
vagrant@debian:~$ systemctl restart ntp
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ===
Authentication is required to restart 'ntp.service'.
Authenticating as: vagrant,,, (vagrant)
Password:
==== AUTHENTICATION COMPLETE ===
vagrant@debian:~$
```

5. Compruebe el servidor NTP está utilizando dichos servidores para la sincronización, y que la hora está sincronizada correctamente

ntpq -p

```
root@debian:/home/vagrant# systemctl restart ntp
root@debian:/home/vagrant# ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
 2.59.235.35 (35 178.215.228.24   3 u   13   64    1   31.038    9.474   0.000
 185.179.104.7   SHM(2)           2 u   12   64    1   44.901   10.256   0.000
 ntp.netnod.se   .PPS.            1 u   16   64    1   75.588   10.008   0.000
 static.135.154. 212.7.1.132      2 u   16   64    1   81.605   12.954   0.000
root@debian:/home/vagrant# date
Sun Apr 23 00:35:09 UTC 2023
root@debian:/home/vagrant#
```